

Owen Peckham MEng

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PhD candidate at the University of Bristol exploring novel techniques for computational design of hollow structures. I have published **7 papers** and presented at **3 conferences**. I am interested in Topology Optimisation, Generative Design, Physics-informed Machine Learning, and Additive Manufacturing.

Present Appointment

PhD Candidate, DMF Lab, The University of Bristol

2023–date

- Third-year PhD candidate in the Design and Manufacturing Futures (DMF) Lab.
- I am assessing the feasibility of the use of hybrid phase-field and level set techniques for hollow structural topology optimisation, as well as novel approaches that maximise second moment of inertia [5].
- Thesis title: “Improving Generative Design: An Investigation Into Strategies For Lightweighting Using Hollow Structures”.
- Worked with others in the DMF on projects related to my main research that address additive manufacturing, topology optimisation and generative design, and AI [2, 4, 6, 7].
- **Scholarship recipient** for the Arabic graduate language training programme **awarded a 2:1**.
- Undertook a 3-month research trip to the Integrated Design Innovation Group at **Carnegie Mellon University**, USA. We collaborated to develop machine learning surrogate models for topology optimisation for use with GPU-based branch and bound optimisers from September to December 2025. Currently finalising the manuscript of an article targeting the Journal of Engineering Design as first author.

Previous Appointments

Research Engineer, NDT Lab, The University of Bristol

Summer 2024–Spring 2025

- Concurrent to my PhD I worked on a project researching low-cost microfluidics produced using additive manufacturing in the Non-Destructive Testing (NDT) Lab.
- Optimised the manufacturing procedure of the microfluidic devices and developed an Autodesk Fusion Computer Aided Design (CAD) add-in tool that automates the design of the microfluidic scaffolds.
- **Invited to deliver a 2-day technical workshop** on microfluidic manufacturing methodologies at the Chulabhorn Research Institute (Bangkok, Thailand).

Research Internship, DMF Lab, The University of Bristol

Summer 2022

- Investigated the impact of information sharing within distributed decentralised agent based additive manufacturing networks [1]. To do this, I used AnyLogic to simulate the network and created a classification system that enabled robust parameterisation of the necessary communication.

Internship, Cambustion Ltd.

Summer 2021

- Designed and built an Automatic Number Plate Recognition (ANPR) device that could be triggered by one of Cambustion’s roadside gas analysers. I implemented the ANPR functionality using a Raspberry Pi 4 with a camera, using APIs from Google (text recognition) and the UK Drivers & Vehicles Standards Agency. The analogue input from the analyser was parsed using an Arduino before the trigger signal was given to the Pi.
- Designed and built a test rig that enabled engineers to test the leakage rate of pressurised fuel injectors. Here I used another Pi 4 for the controller and developed a cloud-linked spreadsheet as a user interface.

Education

The University of Bristol, Undergraduate

2019–2023

- Mechanical and Electrical Engineering, MEng - **awarded first class honours**.
- **Awarded first class honours and joint highest mark in cohort** for my fourth-year group industrial project: “The Canute Braille Reader: Motor Analysis, Control Methods, and Alternative Solutions”. My contribution to this project involved a fundamental redesign of the partner company’s multi-line braille e-reader, evaluating concepts before constructing and performing a feasibility study on the most promising solution.
- **Awarded first class honours** for my third-year individual research project: “An investigation into the use of least-squares system-identification and analytical methods to tune a Proportional-Integral (PI) controller for

a Buck converter with a resistive load”. For this project I devised a method of using least-squares regression to determine the component sizes in a Buck converter. Using this knowledge my automated controller tuner successfully managed the steady-state error and rise times of the converter’s step response to within 10% of the desired values.

The Perse Secondary School and Sixth Form

2012–2019

- Cambridge Pre-U and A-levels - Maths (D2 - A* equivalent), Further Maths (D2), Physics (D3 - A equivalent), Economics (A).
- 10 IGCSEs (and an equivalent) - including Maths, English (Lit and Lang), and the 3 sciences.

Peer-Reviewed Publications

(+) denotes conference papers where I was/ will be the presenting author.

(*) denotes work done in collaboration with international or non-academic co-authors.

Journal

- [7] **O. Peckham**, J. Raines, E. Bulsink, M. Goudswaard, J. Gopsill, D. Barton, A. Nassehi, B. Hicks, (June 2025), “Artificial Intelligence in Generative Design: A Structured Review of Trends and Opportunities in Techniques and Applications”, Journal Designs, Mechanical Engineering Section, [doi:10.3390/designs9040079](https://doi.org/10.3390/designs9040079)
- [6] ***O. Peckham**, C. W. Elverum, B. Hicks, M. Goudswaard, C. Snider, M. Steinert, S. W. Eikevag, (May 2024) “Investigating and Characterising the Systemic Variability When Using Generative Design for Additive Manufacturing”, Applied Sciences, [doi:10.3390/app14114750](https://doi.org/10.3390/app14114750)

Conference

- [5] **+O. Peckham**, B. Hicks, M. Goudswaard (August 2025), “Lightweighting using shells: An exploration of generatively designed hollow structures”, Proceedings of the Design Society, [doi:10.1017/pds.2025.10241](https://doi.org/10.1017/pds.2025.10241)
- [4] ***D. Obilande, O. Peckham**, A. McClenaghan, J. Gopsill, P. Torlind, (May 2024), “Stress Concentrations and Design for Additive Manufacturing: A Design Artefact Approach to Investigation”, Proceedings of the Design Society, [doi:10.1017/pds.2024.181](https://doi.org/10.1017/pds.2024.181)
- [3] **+O. Peckham**, B. Hicks, M. Goudswaard (May 2024), “Are Generative Design Tools Creative? A Characterisation of Tools Throughout the Design Process”, Proceedings of the Design Society, [doi:10.1017/pds.2024.71](https://doi.org/10.1017/pds.2024.71)
- [2] A. McClenaghan, J. Gopsill, **O. Peckham**, B. Hicks (2024), “Examining the Trade-offs in Generating Designs for Many Manufacture”, Procedia CIRP, [doi:10.1016/j.procir.2024.07.061](https://doi.org/10.1016/j.procir.2024.07.061)
- [1] **+O. Peckham**, M. Goudswaard, C. Snider, J. Gopsill (2023), “What to Share? A Preliminary Investigation into the Impact of Information Sharing on Distributed Decentralised Agent-Based Additive Manufacturing Networks”, Advances in Production Management Systems, [doi:10.1007/978-3-031-43666-6_36](https://doi.org/10.1007/978-3-031-43666-6_36)

Teaching

Graduate Teacher: 4th Year Advanced Mechanical Engineering (Generative Design) 2023–date

- **Developed and deliver a guest lecture** each year on the technical specifics of level set topology optimisation.
- The projects for this unit include the design of motorbike components and an extended coursework.

Group Manager & Graduate Teacher: 2nd Year Mechanical Engineering Practice 2023–date

- Technical and managerial advisor to **4 undergraduate design teams of 6 students** each year. I lead the groups through three design projects each year, guiding them through the design process for each.
- The projects for this unit include the design of a jumping robot, quadbike axle and foldable truck-mounted solar panel array.
- As a marker for the unit I conduct vivas for each of the projects and mark the submitted summative reports for **20 students** each year.

Graduate Teacher: 1st Year Principles of Mechanical Engineering

2023–date

- I give technical guidance during lab sessions, drop-ins and lectures, providing design concept critique and practical advice on mechatronics design and manufacture to whoever requests it within the **cohort of 230** each year.
- The project for this unit involves the design and manufacture of a crane mechanism capable of lifting and lowering a mass across the width of a lab bench.

Graduate Teacher: 3rd Year Power Electronics, Machines & Drive Technologies

2023–2025

- As a graduate teacher on this unit I gave technical guidance during lab sessions, drop-ins and lectures. I

provided both theoretical knowledge about power electronic and machine drive design and practical MATLAB and Simulink help to a **cohort of 120 each year**.

- In this unit the students conduct a series of labs before completing a coursework to design the entirety of an electric drivetrain. In the case of the 2024 assessment, this focussed on design of an elevator.

Reviewing

I have reviewed 2 papers relevant to my expertise for Springer Nature's '**Scientific Reports**' — **the third-most cited journal in the world**.

Leadership

Co-Founder, Co-President and Technical Manager, Bristol Racing Society *2021–2023*

- **Co-Founded Bristol Racing**, a society that builds small electric vehicles for endurance races.
- **Co-President** (2021-23) - **re-elected in 2022**.
- **Technical Manager** (2022/23) - I managed the progress of all the sub-teams, advising on design and manufacturing decisions and directing the overall schedule.
- Chassis and mechanical **sub-team lead** (2021/22) - I led the design and manufacture for the sub-team, managing the members and ensuring that progress was maintained.

Achievements

Greenpower Formula 24+

- **Negotiated the university's first** student self-supervised after-hours workshop access scheme.
- Grew the society to **94 paying members**.
- Awarded **£8,800** from funding grants over two years.
- Built partnerships with industry, including the **National Composites Centre**, who provided workshop space, funding, and resources.
- Oversaw a **45 second reduction in lap time** around Castle Combe racetrack from the previous year with the car developed in my second year as Co-President and Technical Manager of Bristol Racing.
- Led Bristol Racing to success, attaining **top 5/25** against teams from for example Jaguar Land Rover and Renishaw.

Publication Awards

- Awarded '**Editor's Choice**' for [7].

Skills

IT and Technical Skills

- **Expert** in Python and MATLAB programming languages
- Previous experience using Arduino, Java and C++ programming languages
- **Advanced** 3D CAD ability on packages Fusion 360, Inventor and SolidWorks.
- **Highly capable** in systems fabrication using equipment ranging from hand tools and DIY electronics to lathes, mills and MIG welders.

Project Management

- I am a practiced manager, adept at leveraging the skills of my team and leading by example.
- My experience with Bristol Racing taught me the value of structured management, using work breakdown plans, Gantt charts and stage gates to stay on track and prepared for potential obstacles.

References available upon request.